IS101 — Introduction to Computing

SOC 101 Grade Book

College of Southern Nevada · Unit 3 — Spreadsheets · Variation 1 · 100 points · 26 tasks

How to use this document: Download the starter workbook from the Project Grader page. Open it in Microsoft Excel 365 and keep this sheet open beside it. Work through each task in order. The Grade Scale on the second sheet tab uses point values based on 300 total points possible: 270+ = A, 240+ = B, 210+ = C, 180+ = D, below 180 = F. When you're done, save the file and upload it to the grader for instant scoring.

Workbook Setup & Formatting Tasks 1–9

Task 1 3 points

Freeze the header row and the student-name column

WHY THIS MATTERS

Freezing keeps headers and student names visible while scrolling through scores.

WHERE TO FIND IT

Click cell B4 (the first score cell below the header and to the right of names) → View tab → Freeze Panes → Freeze Panes.

HOW YOU'LL KNOW IT WORKED

When you scroll down, the header row stays visible. When you scroll right, the name column stays visible.

Task 2 3 points

Add a comment explaining the curve adjustment

WHY THIS MATTERS

Comments let you document decisions without changing the data.

WHERE TO FIND IT

Right-click any cell in the curve or adjustment area → Insert Comment (or New Note) → type your explanation.

HOW YOU'LL KNOW IT WORKED

A red triangle appears in the cell corner; hovering shows your comment.

Task 3 2 points

Format the percentage column with the Percent number format

WHY THIS MATTERS

Percent format displays decimal values as percentages (0.85 becomes 85%).

WHERE TO FIND IT

Select the Percent column data (column J) → Home tab → Number group → Percent Style.

HOW YOU'LL KNOW IT WORKED

Values display as 85% instead of 0.85.

Task 4 3 points

Highlight failing scores with conditional formatting

WHY THIS MATTERS

Conditional formatting visually flags students who need attention. Highlight Total scores (column H) below 180 (the minimum passing score out of 300 points).

WHERE TO FIND IT

Select the Total column data (H4:H11) → Home tab → Conditional Formatting → Highlight Cell Rules → Less Than → enter 180 → choose red fill → OK.

HOW YOU'LL KNOW IT WORKED
Cells with totals below 180 are highlighted in red.

Task 5 3 points

Name the grade-scale lookup range “GradeScale”

WHY THIS MATTERS
Named ranges make formulas easier to read. The GradeScale table is on the second sheet tab and contains point cutoffs and letter grades.

WHERE TO FIND IT
Click the GradeScale sheet tab → select the range A4:B8 (the point values and letter grades, not the headers) → click in the Name Box (left of the formula bar) → type GradeScale → press Enter.

HOW YOU'LL KNOW IT WORKED
The Name Box shows “GradeScale” when the range is selected.

Task 6 3 points

Use the GradeScale named range in a formula

WHY THIS MATTERS
Referencing a named range makes formulas self-documenting.

WHERE TO FIND IT
When you write your VLOOKUP formula (Task 16), type GradeScale instead of the cell range. The grader checks that the word “GradeScale” appears in one of your formulas.

HOW YOU'LL KNOW IT WORKED
The formula bar shows a formula containing “GradeScale” instead of cell references.

Task 7 2 points

Wrap text in the Feedback column

WHY THIS MATTERS
Wrap Text shows long feedback on multiple lines within the cell.

WHERE TO FIND IT
Select the Feedback column (column M) → Home tab → Alignment group → Wrap Text.

HOW YOU'LL KNOW IT WORKED
Long feedback text wraps to multiple lines within each cell.

Task 8 3 points

Convert the grade data to an Excel Table

WHY THIS MATTERS
Excel Tables add automatic filtering, banded rows, and structured references.

WHERE TO FIND IT
Click anywhere in the grade data → press Ctrl+T → verify the range → OK.

HOW YOU'LL KNOW IT WORKED
Filter arrows appear on each column header; the Table Design tab appears.

Task 9 3 points

Sort by Total (descending) then Last Name (ascending)

WHY THIS MATTERS
Multi-level sorting puts the highest scores first and alphabetizes ties.

WHERE TO FIND IT

Data tab → Sort → Sort by: Total → Largest to Smallest → Add Level → Then by: Last Name → A to Z → OK.

HOW YOU'LL KNOW IT WORKED

Students with the highest totals appear first; tied students are alphabetical.

Formulas & Functions Tasks 10–18

Task 10 6 points

Use SUM and AVERAGE to calculate each student's Total and Average

WHY THIS MATTERS

SUM adds all scores; AVERAGE calculates the mean. The Total column is H; the Average column is I.

WHERE TO FIND IT

Click cell H4 → type a SUM formula referencing the score cells (columns C through G) for that student → Enter. Click cell I4 → type an AVERAGE formula referencing the same score cells → Enter. Copy both formulas down for all students.

HOW YOU'LL KNOW IT WORKED

Column H shows each student's total points; column I shows their average score.

Task 11 4 points

Use COUNTA to count submitted assignments per student

WHY THIS MATTERS

COUNTA counts non-empty cells — useful for tracking how many assignments each student turned in. Place the formula in column N.

WHERE TO FIND IT

Click cell N4 → type a COUNTA formula referencing the score cells (C4:G4) for the first student → Enter. Copy down for all students.

HOW YOU'LL KNOW IT WORKED

Column N shows a count (like 5) for each student.

Task 12 4 points

Use an absolute reference ($) in the Percent formula

WHY THIS MATTERS

Absolute references lock a cell so it doesn't shift when you copy. The Percent column (J) divides each student's Total by the Total Points Possible in cell P4. Use $ signs on the P4 reference so it stays locked when copied down.

WHERE TO FIND IT

Click cell J4 → type a formula that divides the Total (H4) by the Total Points Possible, using $ signs on the reference to P4 → Enter. Copy down for all students.

HOW YOU'LL KNOW IT WORKED

The formula bar shows $ signs in the denominator reference. Copying down keeps the same denominator cell.

Task 13 5 points

Use IF to mark each student Pass or Fail

WHY THIS MATTERS

IF tests a condition and returns one value if true, another if false. A student passes if their Total (column H) is 180 or above (60% of 300 points). Place the formula in the Pass/Fail column (K). **Important:** Reference the cell address (like H4), not the word "Total." Complete Task 10 first so column H has values.

WHERE TO FIND IT

Click cell K4 → type an IF formula that checks whether the Total in column H is greater than or equal to 180. If yes, show "Pass"; if no, show "Fail." → Enter. Copy down for all students.

HOW YOU'LL KNOW IT WORKED

Column K shows “Pass” for students with 180+ total points and “Fail” for those below.

Task 14 6 points

Use a nested IF to assign letter grades A–F

WHY THIS MATTERS

A nested IF tests multiple conditions in sequence. Use the Total (column H) and the point cutoffs from the Grade Scale: 270+ = A, 240+ = B, 210+ = C, 180+ = D, below 180 = F. Place the formula in the Letter column (L). Reference cell addresses, not words.

WHERE TO FIND IT

Click cell L4 → type a nested IF formula that checks the Total in column H against each cutoff, starting with the highest (270) and working down → Enter. Copy down.

HOW YOU'LL KNOW IT WORKED

Column L shows a letter grade (A, B, C, D, or F) for each student based on their total points.

Task 15 4 points

Use COUNTIF to count how many students earned an A

WHY THIS MATTERS

COUNTIF counts cells matching a condition — useful for class statistics. Place the formula in the summary area near cell P6. Count the Letter grades in column L.

WHERE TO FIND IT

Click cell P6 (or a nearby empty cell) → type a COUNTIF formula that counts how many cells in the Letter column (L4:L11) contain “A” → Enter.

HOW YOU'LL KNOW IT WORKED

The cell shows the number of students who earned an A.

Task 16 6 points

Use VLOOKUP to look up each letter grade from the Grade Scale

WHY THIS MATTERS

VLOOKUP searches the GradeScale table for each student’s Total and returns the matching letter grade. Reference the Total in column H and use the GradeScale named range (which you created in Task 5). Place the formula in an empty column.

WHERE TO FIND IT

Click an empty cell in the first student’s row → type a VLOOKUP formula that looks up the Total (column H) in the GradeScale named range and returns the letter grade (column 2). Use TRUE for approximate match → Enter. Copy down.

HOW YOU'LL KNOW IT WORKED

Each student shows a letter grade looked up from the Grade Scale table.

Task 17 5 points

Wrap the VLOOKUP in IFERROR to handle errors

WHY THIS MATTERS

IFERROR catches errors (like #N/A) and shows a friendly message instead. Both IFERROR and VLOOKUP must be in the same cell — modify your formula from Task 16.

WHERE TO FIND IT

Edit the VLOOKUP cell from Task 16 → wrap the entire VLOOKUP inside IFERROR, with a fallback message like “Check” → Enter. Copy down.

HOW YOU'LL KNOW IT WORKED

Students with valid totals show letter grades; any errors show “Check” instead of #N/A.

Task 18 3 points

Use TODAY() to stamp the report-generated date

WHY THIS MATTERS

TODAY() inserts the current date and updates automatically. Place it next to the “Report Generated:” label in cell R3.

WHERE TO FIND IT

Click cell S3 (next to “Report Generated:”) → type =TODAY() → Enter.

HOW YOU'LL KNOW IT WORKED

The cell shows today’s date.

Charts Tasks 19–21

Task 19 4 points

Create a Column chart of class average by assignment

WHY THIS MATTERS

Charts visualize how the class performed on each assignment.

WHERE TO FIND IT

Select the assignment headers (C3:G3) and the class averages for those assignments → Insert tab → Charts group → Column → Clustered Column.

HOW YOU'LL KNOW IT WORKED

A column chart appears showing average scores for each assignment.

Task 20 4 points

Add data labels and a chart title

WHY THIS MATTERS

Data labels show exact values on each bar; a chart title identifies the chart.

WHERE TO FIND IT

Click the chart → Chart Design tab → Add Chart Element → Data Labels → Outside End. Then Add Chart Element → Chart Title → type a descriptive title.

HOW YOU'LL KNOW IT WORKED

Each bar shows its value; the chart has a title at the top.

Task 21 3 points

Add alt text to the chart

WHY THIS MATTERS

Alt text describes the chart for screen readers and accessibility.

WHERE TO FIND IT

Right-click the chart → Edit Alt Text → type a description of what the chart shows.

HOW YOU'LL KNOW IT WORKED

Right-clicking the chart and selecting Edit Alt Text shows your description.

Finishing Touches Tasks 22–26

Task 22 3 points

Bold and shade the header row

WHY THIS MATTERS

Bold text and a colored background make headers stand out.

WHERE TO FIND IT

Select the header row (row 3) → Home tab → Bold (Ctrl+B) → Fill Color → choose a color.

HOW YOU'LL KNOW IT WORKED
The header row has bold text on a colored background.

Task 23 4 points

Add a header with the course name and a footer with page number

WHY THIS MATTERS
Headers and footers identify printed pages.

WHERE TO FIND IT
Page Layout tab → Page Setup dialog launcher → Header/Footer tab → Custom Header: type “SOC 101 Grade Book” → Custom Footer: insert page number.

HOW YOU'LL KNOW IT WORKED
Print Preview shows the course name at the top and page number at the bottom.

Task 24 3 points

Set Title and Author workbook properties

WHY THIS MATTERS
Metadata makes the workbook identifiable and searchable.

WHERE TO FIND IT
File tab → Info → Properties panel. Set Title to “SOC 101 Grade Book” and Author to your name.

HOW YOU'LL KNOW IT WORKED
File {ARR} Info shows Title and Author populated.

Task 25 3 points

Filter to students currently failing

WHY THIS MATTERS
Filtering shows only students who need intervention. Filter the Letter column (L) to show only “F” grades.

WHERE TO FIND IT
Click the dropdown arrow on the Letter column header → uncheck Select All → check only “F” → OK.

HOW YOU'LL KNOW IT WORKED
Only students with an F grade are visible.

Task 26 8 points

Center the score columns

WHY THIS MATTERS
Centering aligns numeric scores in the middle of each cell for a clean look.

WHERE TO FIND IT
Select all score columns (C through G) → Home tab → Alignment group → Center.

HOW YOU'LL KNOW IT WORKED
All score values are centered in their cells.

Total: 100 points. Every task is graded automatically when you upload your file. © Dr. Mark Taormino · All Rights Reserved · College of Southern Nevada